Do Now · retrieval

CPU components and registers

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which data structure supports last-in-first-out Undo?

Q2

What is the extra space complexity of allocating a new length-n array?

Q3

What does the ALU perform?

Q4

Simplify (A ∧ ¬B) ∨ (A ∧ B) and explain why the grouped cells make one variable disappear.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand the CPU areas — The control unit decodes and coordinates operations; the ALU performs arithmetic and logical operations.

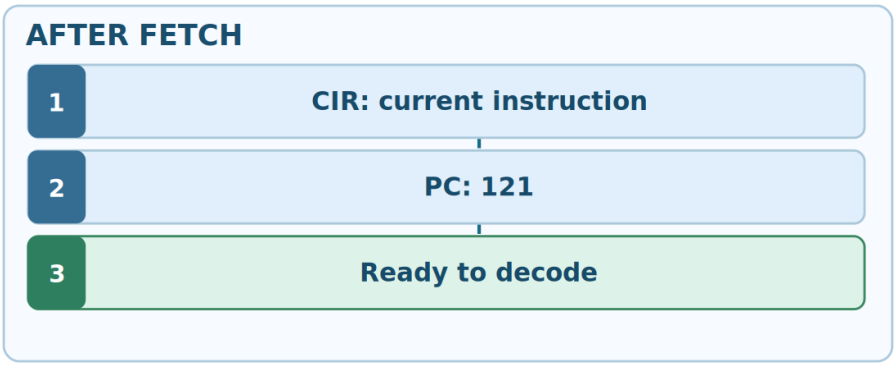
Core idea 2

Separate address from data — MAR holds the address being accessed; MDR holds the instruction or data transferred; PC holds the next instruction's address.

Core idea 3

Trace the simplified fetch — With PC=120, MAR becomes 120; memory supplies instruction via MDR to CIR; PC advances to the next sequential address. Branching may change PC later.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

ALU — Arithmetic Logic Unit.

CU — Control Unit, coordinates and decodes instruction operations.

MAR — Memory Address Register.

MDR — Memory Data Register.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

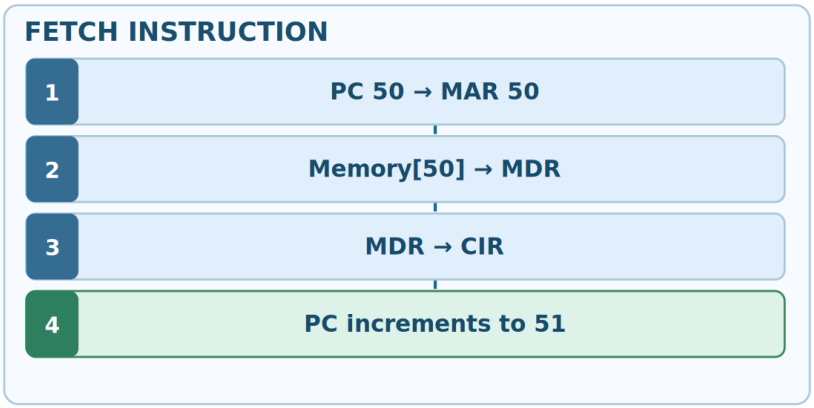
Problem

If PC starts at address 50, trace register roles for the next sequential instruction fetch.

Follow the worked method

1. Predict first — Copy PC=50 into MAR to identify the memory address being accessed.
2. Explain the answer — Fetch memory content into MDR, copy it to CIR and advance PC to 51.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
MAR must contain 200 and MDR transports the instruction from memory.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State the roles of PC, MAR, MDR, CIR and ACC.

Question 2

Trace a fetch starting PC=73 where Memory[73] contains LDA 92. State each register's role and the sequential next PC.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why the CPU uses both MAR and MDR rather than a single undifferentiated register, referring to an instruction fetch and a data load.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The MAR stores the contents of the memory location being fetched.

Use key vocabulary from memory

1. Explain ALU in your own words.

2. Explain CU in your own words.

3. Explain MAR in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.